

Kushagra Srivastava

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Education

Arizona State University, BS/MS in Computer Science (4+1)

August 2022 – May 2027

- *Relevant Coursework:* Data Structures and Algorithms, Software Engineering, Databases, Theory of Computation, Artificial Intelligence, Linear Algebra

Skills

Languages: Python, JavaScript/TypeScript, Java, C++, Swift, SQL

Frameworks and Libraries: FastAPI, Next.js, Node.js, Flask, React, D3.js, SQLAlchemy, Celery

Tools: GCP (Cloud Run, Vertex AI), Firebase, **Docker**, Redis, **PostgreSQL**, Git, Linux/UNIX, **CI/CD**, JSON schema validation, telemetry logging

Experience

Backend Software Engineer, SkipCourse LLC

Aug 2025 – Present

- Achieved **99% backend reliability** serving **500+ daily requests** by engineering a **cloud-native GCP** service with **distributed retry logic** and comprehensive error handling.
- Reduced **Gemini inference** auth failures to zero and maintained **p95 latency under 450ms** by re-architecting the pipeline with an authenticated **Flask** gateway and **exponential backoff**.
- Reduced **error rate by 17%** by deploying **request telemetry** and error dashboards that tracked p95 and failure thresholds for data-driven iterations.
- Secured **10K+ monthly API calls** by auditing **IAM policies** to enforce least-privilege access across **Vertex AI** production inference.

Frontend and Web Performance Engineering Intern, GRID

Jul 2025 – Aug 2025

- Improved page-load performance by **25%** across **15K+ monthly visits** by implementing **bundle optimization**, lazy loading, and DOM restructuring, reducing TTI from 3.2s to 2.4s.
- Saved **8 hours per week** of editorial overhead by building **admin-configurable content modules** that cut manual publishing effort by 40%.
- Increased **user retention by 18%** by configuring an **analytics pipeline** and partnering with stakeholders to prioritize data-driven UX improvements.

Student Data Verifier, Admission Services, Arizona State University

Oct 2023 – Jun 2024

- Maintained **99.7% data integrity** across **5K+ applicant records** by executing **FERPA-compliant** verification workflows spanning PeopleSoft, Salesforce, and document repositories.
- Reduced processing time from **12 to 9 minutes per application** and cut verification errors by 15% by building **Excel dashboards** that tracked throughput and error rates.

Project Work

TOURGAIDE – AI-Powered Travel Platform | Python, FastAPI, PostgreSQL, Redis, Celery, Docker

- Achieved **98%+ data validation accuracy** on **1,200+ daily API calls** by building an **async AI itinerary engine** with structured schema enforcement and sub-300ms latency.
- Improved request success rate from **85% to 98%** and cut monthly API costs by **USD 240** by implementing **Redis caching** and intelligent request routing.

PYROCACHE – AI-Augmented In-Memory Cache Engine | Python, asyncio, Vertex AI, Docker

- Engineered a **Redis-compatible cache server** from raw TCP sockets in Python – custom **RESP** protocol parser, **asyncio** event loop, and command dispatcher supporting GET, SET, DEL, TTL, SUBSCRIBE, and PUBLISH.
- Reduced **cache miss rate by 34%** by building a **Vertex AI** key prefetcher that analyzes access patterns and pre-warms predicted hot keys before request arrival.
- Achieved **sub-millisecond p99 latency** at **50K+ ops/sec** via single-threaded async I/O, **LRU/LFU eviction**, min-heap **TTL scheduling**, and **AOF persistence** for zero data loss on restart.

Awards and Certifications

- **Claude with Google Vertex AI** – Anthropic: certified in **Vertex AI inference**, **production AI deployment**, and **IAM security** on GCP (March 2026)
- **AI Fluency: Framework and Foundations** – Anthropic: applied AI concepts, **model deployment workflows**, responsible AI practices (2026)
- **New American University Scholarship** (ASU): merit-based renewable scholarship, academic excellence